Asymmetric reserve allocation in the affine local-moment bound

Zhongpai Gao
Prepared with AI assistance

13 October 2026

Abstract

We refine the affine local-moment argument to the exact boundary $8748673/10000000 = 0.8748673$. The change allocates the available LOW saving unequally between auxiliary fees and retained saving, enlarging the permitted capacity defect to $1/241$. The amplifier, source-volume cap, and final analytic savings are retained. Exact row certificates then give the revised endpoint. The full analytic argument builds on Kari Roegnvaldsson's affine capacity, nielstron's local $22/5$ proof, and their mathematical predecessors.

Theorem 1. *For every positive integer q, every complex Dirichlet character χ modulo q, and every $s \in \mathbb{C}$ satisfying*

$$\Re s > \frac{8748673}{10000000}, \qquad \neg(\chi = 1 \text{ and } s = 1),$$

the canonical analytically continued Dirichlet L-function $L(\chi, s)$ is nonzero.

1 The saving available to the actual source

The affine local argument retains two separate geometric requirements:

$$n_1 + n_2 + z \le \frac{881}{1000}, \qquad n_1 + n_2 + \frac{9}{2}z \le 1 + g.$$

Here z is the local prime length and g bounds the old-capacity defect. The amplifier remains $a = 3/500$. The raw LOW reserve supplied by the incumbent ledger is

$$r_{\mathrm{raw}}(g) = \frac{3}{500} - \frac{67}{450}\left(g + 6\frac{3}{500}\right).$$

This is a reserve for the original coefficient source, before the separately recorded auxiliary losses. At $g = 1/241$,

$$r_{\mathrm{raw}}(1/241) = \frac{301}{13556250} > \frac{1}{46000}.$$

Use the following unequal split:

Raw reserve	Paid auxiliary fees	Retained LOW saving
$1/46000$	$1/2000000$	$1/48000$

It is valid because

$$\frac{1}{46000} - \frac{1}{2000000} - \frac{1}{48000} = \frac{7}{17250000} > 0.$$

Thus, whenever the actual padded-source ledger has fee pad+loss+$(67/450)$ geometryLoss $\le 1/2000000$, its exponent is at most the nominal target minus $1/48000$. The fee bound and retained saving are distinct quantities in the source inequality; there is no requirement to divide the raw reserve into equal halves.

The common tiny loss is chosen to be $\eta = 10^{-8}$. The original fixed profile and logarithmic losses can be made this small by enlarging the common height threshold. After eleven such fees, the retained saving still exceeds the unchanged final LOW target:

$$\frac{1}{48000} - 11\eta - \frac{1}{50000} = \frac{217}{300000000} > 0.$$

For the extracted errors, the same annular amplifier gives prime length $\lambda \geq 9/2000$. The three raw reserves remain at least $1/1250$:

$$\lambda - \frac{g + 9/500}{6}, \qquad 6\lambda - \frac{g + 9/500 + 6\lambda}{6}, \qquad 7\lambda - \frac{g + 9/500 + 6\lambda}{6} \;\geq\; \frac{1}{1250}.$$

The first, tightest expression is $1169/1446000 > 1/1250$ at the endpoint. This retains the downstream $1/2000$, $1/2500$, and final $1/10000$ savings.

2 Capacity and coefficient transport

For the same bare physical window $419/1000 \leq n \leq 422/1000$, define

$$b_g(n) = \frac{1 + g - 2n}{9/2}, \qquad c_g(n,t) = \max\left\{0, \frac{1 + g - 2(n+t)}{9/2}\right\}.$$

The local branch is used only when $0 \leq t \leq 3/4000$. With $g = 1/241$, the original volume gate follows from the exact worst-case bound

$$2(n+t) + c_g(n,t) \leq \frac{3820717}{4338000} < \frac{881}{1000}.$$

Also $c_g(n,t) \leq 20021/542250 < 37/1000$ and $n + t < 423/1000$, so the original physical window remains valid. The two diagonal constraints for independent plain heights are carried through separately.

Past $t = 3/4000$, use the old capacity $c_0(n,t) = \max\{0, (1 - 2(n+t))/(9/2)\}$. If $c(n,t)$ denotes the resulting piecewise capacity, then

$$b_g(n) - c(n,t) \leq 2t.$$

On the local branch this follows directly from coefficient dilation. On the old branch it follows from $g \leq 7t$, since $g = 1/241 < 21/4000$. For the actual scalar parameters $0 \leq q \leq \delta/2$, this gives the repayment

$$\frac{q\, b_g(n)}{2} \leq \frac{\delta t}{2} + \frac{q\, c(n,t)}{2}.$$

Consequently the wider capacity is paid by the original weighted coefficient mass, including free shifts and the explicitly charged profile padding.

3 Exact geometry and the complete row saving

The linked parameters are

$$\theta = \frac{8748673}{10000000}, \qquad \ell = \frac{1253981}{7500000}, \qquad h = \frac{397}{500},$$
$$\kappa_0 = \frac{3748673}{5000000}, \qquad y = h - 2\ell = \frac{1723519}{3750000}, \qquad x = 1 + \ell - h = \frac{2798981}{7500000}.$$

They satisfy $\theta = 11/12 - \ell/4$ and $\kappa_0 = 2\theta - 1$. Set $j = h - 1/1000$ and $\zeta = 10^{-10}$. For $0 \leq \delta \leq 3/4$ and $0 \leq p \leq 1/2$, put

$$V = 5/6 - \delta, \quad D = 3 - p - \frac{2p}{3\kappa_0}, \quad B = 2(1-p)\left(1 - \frac{p}{3\kappa_0}\right), \quad J = VD + \delta B > 0.$$

The literal row-count exponent is $R_{\kappa_0} = 1 - \delta + \delta BV/(2J)$. Above the junction, the normalized affine-source exponent is

$$E_g(d,\delta,p) = \frac{1}{2} + \frac{13h}{75} - \frac{y}{2} - \frac{\ell}{2} + \frac{33d}{50} + \delta\left(\frac{1}{2} - \frac{3d}{4}\right) + \delta p\left(\ell - \frac{2y - d + gd}{9}\right) + \frac{3d\delta BV}{8J}.$$

The added gd term is the retained affine defect allowance, expressed in the actual source ledger. Below the junction the incumbent 6κ source estimate is retained, with its original breakpoint at $d = y$.

Multiplying $\theta - r - E_g$ by $8J > 0$ and putting $u = 1/2 - p$ gives $Q_0(\delta) + uQ_1(\delta) + u^2Q_2(\delta) + u^3Q_3(\delta)$. The coefficients are regenerated exactly. For $i = 0, 1$, writing $Q_i(t) = a_it^2 + b_it + c_i$, one checks $a_i > 0$ and $4a_ic_i - b_i^2 \geq 0$. Then $4a_iQ_i(t) = (2a_it + b_i)^2 + (4a_ic_i - b_i^2) \geq 0$. The remaining two quadratics have nonnegative coefficients on $t \geq 0$. The top row's first discriminant gap is, exactly,

$$\frac{340952468857528755695910111111}{3012185863961537158512562500000000000} > 0.$$

The same regenerated certificate method proves reserves $1/2000000$ at $d = h$, $1/10000$ at $d = j$, and $1/100$ at $d = 1/2$. The full coefficient tables and exact identities accompany the source.

Interpolation covers $1/200 \leq d \leq h + \zeta$. The top slope is at most 1, and $1/2000000 - 2\zeta > 1/3000000$ pays its extension. The inherited κ-perturbation estimate, applied with the actual $\kappa = 2\beta - 1$, therefore gives the complete row saving

$$G_{2\beta-1,g}(d, \delta, p) - \beta \leq -\frac{1}{3000000} - \frac{3}{4}(\beta - \theta) \qquad (\beta \geq \theta). \tag{1}$$

Here G uses the affine ledger above j and the incumbent ledger below j. The additional conservative transport certificate uses anchor $874867/1000000 \leq \theta$, $\ell \in (167197, 167198)/10^6$, and $y > 45960/100000$. Its positive rational Bernstein expansion retains the separate reserve $1/2000$; all twelve coefficients and its exact identity are included with the source.

4 The unconditional endpoint

The local moment and the geometric reserve checks are applied to the original source packet. The complete analytic proof retains coherent amplification, joint row reconstruction, cancellation in the first-kernel errors, coprimality, ramified labels, finite Euler deletions, independent heights, and the low, exceptional, and large-height estimates. The actual padded-source consumers use the asymmetric split in Section 1. The common auxiliary losses and height are chosen before the target character.

The literal physical exponent is identified with its scalar ledger and actual normalization. Equation (1) supplies a positive saving; integration and normalization prove the required analytic source bound. The same normalized probe appears in LOW and HIGH. The retained analytic correction and principal-term estimates then complete the continuation argument, bounding the actual Hecke-zero supremum by θ. Finally, the proved Hecke-to-Dirichlet identity yields every modulus and character in Theorem 1, with the sole principal pole excluded. The Lean endpoint is the canonical theorem `Candidate.bound`.

Attribution and sources. This reserve allocation and endpoint adaptation are by Zhongpai Gao, with AI assistance. Kari Roegnvaldsson introduced the retained affine capacity in PR 37 and the preceding reserve reallocation in PR 38. The complete local 22/5 analytic proof is nielstron's PR 26. Their predecessors include OpenAI's quasi-Riemann development, Akash Levy's weighted numerator, Hailey Collet's Cycle25 development, and Gao's earlier local-moment adapters. Full source pins and original notices are retained in the source package and accompanying README, under Apache-2.0 and the dependencies' original licenses.